



BCS 753

RAG-Augmented Speech-to-ISL Avatar Generator

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Introduction

Communication between the Deaf and Hard-of-Hearing community and the hearing population remains a significant accessibility challenge in India. Most sign-language systems handle Sign → Text; the reverse direction — what a hearing person needs to communicate outward — is comparatively underexplored.

Established: Sign → Text

Recognizing signed input from video (CNNs/LSTMs) and converting it to text — a substantial body of prior work, mostly for captioning and recognition.



Underexplored: Speech/Text → Sign

Generating correctly-ordered ISL output from spoken or typed language, so a hearing person can be understood without an interpreter present.

Project Focus *Build the generation direction as an assistive aid — not a replacement for certified human interpreters.*

Problem Statement

Direction Is Reversed

Research has concentrated on recognition, not generation — a substantially different, less-studied problem.

Data Is Scarce

ISL-CSLTR and CISLR are small corpora, limiting how well a generator can be fine-tuned to generalize alone.

Grammar Mismatch

ISL uses topic - comment order. Word-for-word substitution produces ungrammatical gloss.

Net Effect *These compound: a reversed, data-scarce, structurally-divergent problem is exactly where fine-tuning alone defaults back to source-language word order — the failure this project's retrieval-augmented design counters.*

Literature Review

Prior studies establish the major components of speech-to-ISL translation, sign generation, retrieval, resources, and avatar production.

Research Paper	What It Does	Key Limitation
Reddy et al. (2023) [1] ISL generation from live Tamil audio/text	End-to-end generation of ISL from live Tamil speech or text.	Language/domain specific; no retrieval grounding or broad benchmark.
Nair et al. (2023) [2] English → ISL gloss seq2seq	Supervised sequence-to-sequence generation of ISL gloss sequences from English.	Requires parallel gloss data; no inference-time evidence retrieval.
Joshi et al. (2023) [3] ISLTranslate	31K continuous ISL–English sentence/phrase pairs and translation benchmarks.	Primarily supports sign/video → text; not speech/text → ISL RAG generation.
Baltatzis et al. (2024) [4] Neural Sign Actors	Diffusion and anatomically informed 3D representations for realistic sign production.	Non-ISL resources; computationally intensive; retrieval absent.
Yin et al. (2024) [5] T2S-GPT	Autoregressive text-to-sign production using dynamic vector quantization.	German Sign Language focus; does not address low-resource ISL.
Zuo et al. (2024) [6] Spoken2Sign	Retrieve-then-connect architecture for 3D avatar output.	Retrieval is gloss-to-3D-sign lookup, not contextual sentence–gloss grounding; mainly non-ISL.
Joshi et al. (2024) [7] iSign	118K+ ISL–English video/sentence pairs and multiple processing tasks.	Large benchmark, but no matched RAG-on vs. RAG-off generation study.
Varanasi et al. (2024) [8] Linguistically informed ASL Transformers	Injects linguistic information into text-to-ASL gloss translation.	ASL/banking-domain focus; no retrieval component.

Literature Review (contd.)

Research Paper	What It Does	Key Limitation
Merx et al. (2024) [9] RAG for low-resource MT	Retrieved examples and dictionary entries can improve some low-resource MT settings.	Spoken-language MT rather than sign-language gloss/avatar generation.
Dal Bianco et al. (2025) [10] LLM data augmentation	Uses GPT-generated paraphrases to augment sign-language translation training data.	Augmentation is not inference-time sentence–gloss retrieval; possible paraphrase noise.
Inan et al. (2025) [11] SignAlignLM	Integrates sign-language knowledge into text-based and multimodal LLMs.	Broad multimodal scope; not an ISL-specific, leakage-controlled retrieval study.
Imai et al. (2025) [12] SiLVERScore	Introduces semantically aware evaluation beyond conventional back-translation.	Metric does not address ISL data scarcity or the proposed RAG comparison.
Yazdani et al. (2025) [13] VLM-assisted SLT	Uses vision-language modeling to support sign-language translation workflows.	Primarily sign-video → text; does not address speech/text → ISL production.
Yazdani et al. (2025) [14] Continual multilingual SLT	Studies incremental and multilingual learning for sign-language translation.	Resource imbalance/forgetting remain concerns; no sentence–gloss RAG ablation.
Patra et al. (2026) [15] Gloss-driven ISL production	Explores learned pose representations for gloss-driven ISL avatar production.	Depends on gloss/pose coverage; no retrieval-grounded ordering comparison with a matched baseline.

Identified Gap *Sentence-level retrieval-grounded ISL gloss generation remains underexplored, particularly with leakage-controlled RAG-on/RAG-off comparison, structure-aware evaluation, and connection to avatar rendering.*

Proposed Solution

The system grounds gloss generation in similar, already-correct sentence-gloss examples retrieved at inference time, rather than relying solely on a small fine-tuning corpus.

Fine-tuning Alone

A generator fine-tuned only on ISL-CSLTR/CISLR must generalize from a small corpus. With limited data it tends to fall back on source-language word order.



Retrieval-augmented (RAG)

The generator is additionally shown 2–5 retrieved sentence-gloss pairs as few-shot evidence — a concrete ordering pattern to follow.

- The retrieval KB can grow by adding pairs alone — no retraining required.
- If no close match exists, it falls back to zero-shot generation.

Core Research Question *Does retrieval improve gloss generation in a low-resource ISL setting?*

Illustrative Example — Retrieval in Action

Input Sentence *“I am going to school today.”*

- **Top Retrieved Sentence-gloss Pairs**

- “I am going to market today.” → TODAY I GO MARKET (similarity 0.89)
- “She is going to school.” → SHE GO SCHOOL (similarity 0.81)

RAG ON — with context

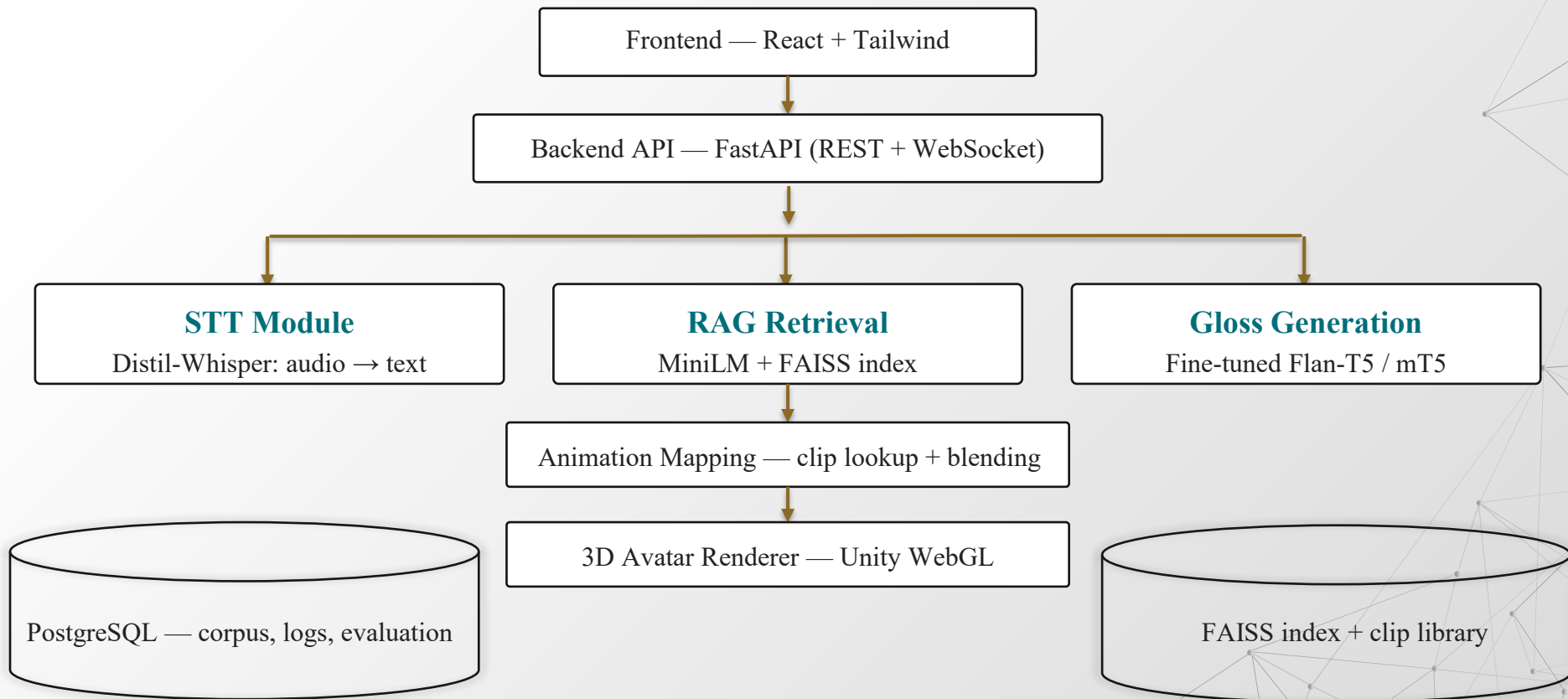
Output: TODAY I GO SCHOOL
BLEU vs. reference: 1.00

RAG OFF — zero-shot

Output: I GO TO SCHOOL TODAY
BLEU vs. reference: 0.62

Reading The Result *Retrieved examples supply the correct topic-comment pattern (time marker first). Without that context, the zero-shot model defaults toward English word order — fluent English, but not well-formed ISL gloss.*

System Architecture



Datasets & Retrieval Knowledge Base

ISL-CSLTR

A published ISL sentence-gloss corpus; one of two base sources for fine-tuning and retrieval.

CISLR

A second sentence-gloss corpus, widening vocabulary and sentence-pattern coverage.

Provenance & Split

- Every entry is tagged with its source (ISL-CSLTR / CISLR / curated) for provenance.
- Split into train / validation / test — test is held out from fine-tuning and the retrieval index, avoiding leakage.

Why This Matters For Rag Specifically *Because the retrieval index is decoupled from the generator's weights, the knowledge base can grow — new pairs re-embedded into FAISS — without retraining the generator at all.*

Technology Stack

Layer	Technology	Role
Frontend	React + Tailwind CSS	Text/microphone input, avatar viewport
Backend	FastAPI (Python)	Orchestrates the pipeline; REST + WebSocket
Speech recognition	Distil-Whisper	Converts spoken audio to text
Embeddings	sentence-transformers (MiniLM)	Encodes sentences for similarity search
Vector index	FAISS	Retrieves the closest sentence-gloss pairs
Gloss generator	Fine-tuned Flan-T5 / mT5	Produces the ISL gloss sequence
Avatar engine	Unity	Renders the avatar, plays clips
Database	PostgreSQL	Corpus, clips, sessions, logs, results
Deployment	Docker	Reproducible, containerized setup

Methodology

01 Speech Transcription

Distil-Whisper converts speech to text; typed input shares the same pipeline.

02 Language Normalization

Text is routed through English, the language retrieval/generation align to.

03 Contextual Retrieval

The sentence is embedded (MiniLM) and matched in FAISS for similar pairs.

04 Gloss Generation

Flan-T5/mT5, prompted with retrieved examples, produces ISL-ordered gloss.

05 Animation Mapping

Each gloss token resolves to a clip; blending smooths the joins.

06 Rendering & Evaluation

The clip sequence plays on the avatar; supports RAG on/off comparison.

Design Principle *Each stage is independently testable — the retrieval index, generator, and animation layer can each be swapped or improved without touching the others.*

Evaluation Plan

Test Levels

- Unit tests per module (STT, retrieval, generation, mapping, API).
- Integration tests across the pipeline, incl. WebSocket streaming.
- System tests: 20+ sentences, cross-browser, Docker reproducibility.

The Core Evaluation: Rag On Vs. Off

- Hold out a test split never used in fine-tuning/retrieval.
- Generate each sentence twice — RAG on, then off.
- Score with BLEU / exact-match plus qualitative ratings.

Success Criteria

100–300

core sentences signed
end to end

<3–5s

latency to
animation start

RAG >no-RAG

measurable quality
gain

Why Bleu Alone Isn't Enough *BLEU is a weak proxy for sign grammaticality, so qualitative fluency ratings from a small reviewer panel are collected too.*

No External Baseline Exists *No published RAG-for-sign-language baseline exists — the project's own on/off ablation is the primary evidence.*

Scope & Future Scope

1. Scope

- 01 **Speech-to-Text** Accept Hindi speech or typed input and convert speech to text.
 - 02 **RAG-based Retrieval** Retrieve similar sentence–gloss examples from the ISL corpus before generation.
 - 03 **Gloss Generation** Generate an ISL-ordered gloss sequence using the fine-tuned model.
 - 04 **Gloss-to-Animation** Map gloss tokens to animation clips and blend the sequence smoothly.
 - 05 **Web-based Demo** Provide a simple web interface for input and 3D avatar output.
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2. Future Scope

- 01 **Hindi-first Expansion**
- 02 **Communication Integration**
- 03 **Non-manual Markers**
- 04 **Mobile Application**

Conclusion

This project bridges spoken language and Indian Sign Language by combining retrieval, language models, and real-time avatar animation — an assistive communication aid and research demonstration, not a replacement for certified human interpreters.

The central research contribution is using retrieved sentence-gloss examples to ground generation under low-resource ISL conditions — evidenced by the RAG-on vs. RAG-off comparison.

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The background of the slide is a light gray with a subtle, abstract pattern of thin black lines and small dots, resembling a network or a molecular structure. The pattern is more dense on the left and right sides, with some lines extending towards the center.

Thank You !
